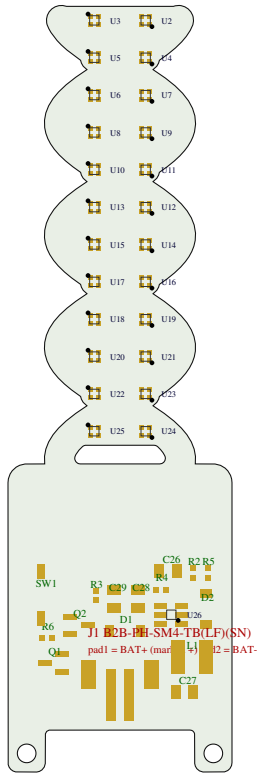
DNA LED pendant - TOP side assembly drawing

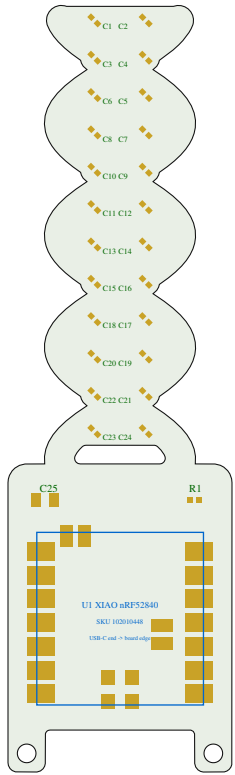
PCB 24.00 x 82.09 mm, 2 layers, 1.6 mm.

Black dot beside an LED = its pin 1 side. Designators are NOT printed on the board silkscreen.



DNA LED pendant - BOTTOM side assembly drawing

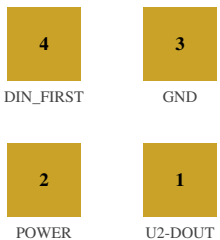
Viewed from below (mirrored). PCB 24.00 x 82.09 mm, 2 layers, 1.6 mm.
Designators are NOT printed on the board silkscreen.



Detail A - WS2812B-type LED XL-1010RGBC (U2..U25), top

TOP side; scale 60:1

Part U2: XL-1010RGBC-WS2812B (SMD4_XL-1010RGBC_XLG) at X=102.75 Y=70.00 rot 180

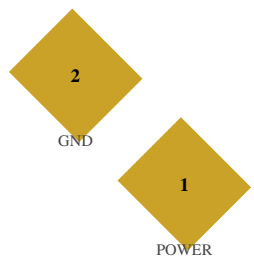


Pad 1 = DOUT, pad 2 = VDD (5 V), pad 3 = GND, pad 4 = DIN. Pin 1 corner is marked by the black dot on the overview.
All 24 LEDs sit on the TOP side at 0/180 degrees; chain order U2 -> U25 (DOUT -> next DIN).

Detail B - bypass capacitor behind each LED (C1 behind U2), bottom

BOTTOM side - viewed from below (mirrored); scale 60:1

Part C1: 100nF (C_0402_1005Metric) at X=102.75 Y=70.00 rot 45

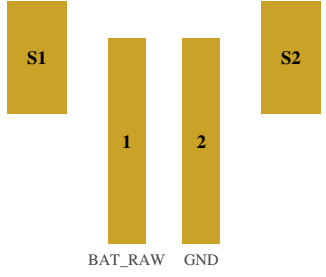


C1..C24: 100 nF 0402, non-polarized, one per LED, on the BOTTOM side exactly behind the LED (rotated 45/-135 degrees).

Detail C - battery connector J1 (JST B2B-PH-SM4-TB), top

TOP side; scale 14:1

Part J1: B2B-PH-SM4-TB (JST_B2B-PH-SM4-TB) at X=100.00 Y=140.00 rot 0

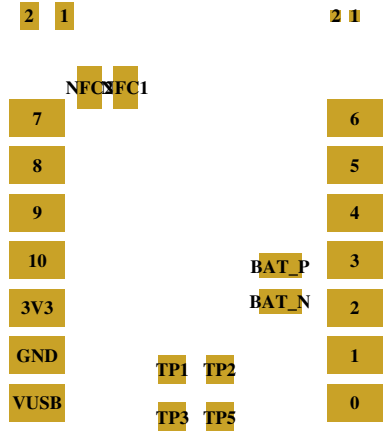


Pad 1 (left, silkscreen +) = BAT+ ; pad 2 = BAT-. Mechanical tabs S1/S2 are soldered but carry no net.

Detail D - XIAO nRF52840 U1, R1 100R and C25 10uF, bottom

BOTTOM side - viewed from below (mirrored); scale 7:1

Part C25: 10uF (C_0805_2012Metric) at X=108.00 Y=121.40 rot 0
Part R1: 100 (R_0402_1005Metric) at X=92.00 Y=121.40 rot 0
Part U1: 102010448 (MODULE_102010448) at X=100.00 Y=134.50 rot 0

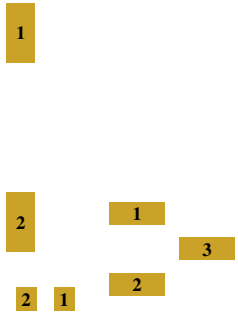


U1: castellated SMD module (Seeed SKU 102010448) placed on the BOTTOM; USB-C connector faces the bottom board edge.
R1 (100 ohm 0402) and C25 (10 uF 0805) are non-polarized.

Detail F - power switch SW1, Q2 and R6, top

TOP side; scale 14:1

Part Q2: AO3401A (SOT-23) at X=95.60 Y=134.80 rot 0
Part R6: 100k (R_0402_1005Metric) at X=92.20 Y=136.15 rot 180
Part SW1: SLIDE_SW (SW_DIP_SPSTx01_Slide_Copal_CHS-01A_W5.08mm_P1.27mm_JPin) at X=91.55 Y=131.55 rot -90

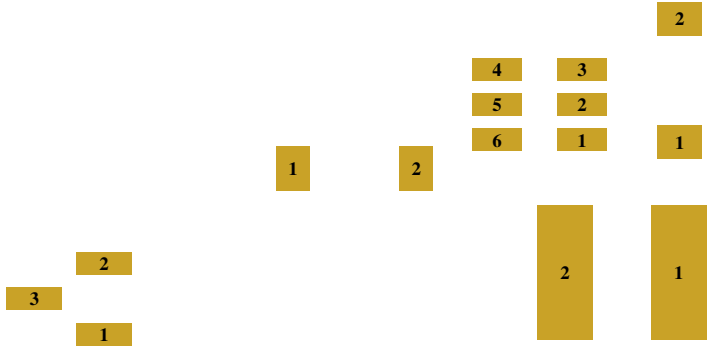


SW1 (Nidec Copal CHS-01A, SMD slide switch): no polarity; ON is the end marked ON on the body. Q2 SOT-23: 1 gate, 2 source (battery side), 3 drain (VBAT after the switch).
R6 (100 kohm 0402) is non-polarized.

Detail E - battery boost U26 (MT3608B), L1, D1, D2, Q1, top

TOP side; scale 14:1

Part D1: 1N5819HW (D_SOD-123) at X=100.55 Y=135.35 rot 0
Part D2: 1N5819HW (D_SOD-123) at X=109.25 Y=133.00 rot 90
Part L1: 10uH (L_Bourns-SRN4018) at X=107.70 Y=138.15 rot 180
Part Q1: 2N7002 (SOT-23) at X=92.90 Y=138.85 rot 180
Part U26: MT3608B (SOT-23-6) at X=105.50 Y=133.65 rot 180



U26 SOT-23-6: pad 1 = SW is on the side with three pads 1-2-3 (pad numbers are drawn). Pins: 1 SW, 2 GND, 3 FB, 4 EN, 5 VIN, 6 NC.
D1 and D2: cathode (bar on the part) at pad 1. Q1 SOT-23: 1 gate, 2 source, 3 drain. L1 non-polarized.
C26-C29 (10 uF 0805) and R2-R5 (0402) are non-polarized; C25..C29 are all the 10 uF part.